



Autism and Employment Challenges: The Double Empathy Problem and Perceptions of an Autistic Employee in the Workplace

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Abstract

Background: High unemployment rates are found among autistic adults. Difficulties with social functioning in non-autistic workplaces can be significant barriers to employment success. Autistic social functioning challenges in non-autistic spaces have traditionally been attributed to assumed impairments in Theory of Mind (ToM). Alternatively, the Double Empathy Problem (DEP) posits that autistic social challenges arise not from assumed impairments within the autistic person but instead from mutual misunderstandings in the autistic/non-autistic social interaction. The purpose of this study was to compare the ToM impairments explanation of autistic social functioning with the DEP, within the context of autistic employee social functioning in a non-autistic workplace.

Methods: This study compared autistic and non-autistic participants' ability to accurately interpret the behaviors of an autistic employee at work. A sample of 254 participants (173 non-autistic and 81 autistic) read a vignette about a hypothetical autistic employee having difficulty coping in the workplace. Participants answered open-ended questions regarding their interpretation of the employee's behavior and emotional state.

Results: A significantly greater proportion of autistic participants (50.7%) accurately interpreted the behavior of the employee compared with non-autistic participants (31.2%) ($\chi^2 = 8.65$, $p = 0.003$). Autistic participants with the highest behavior interpretation scores had significantly higher mean self-reported autism traits scores ($M = 26.8$) compared with autistic participants who scored lowest on behavior interpretation ($M = 19.3$, $p < 0.001$). The opposite relationship was found for non-autistic participants.

Conclusions: Results from this study contribute to evidence supporting the DEP, shifting the paradigm of autistic social functioning away from a deficit model and toward addressing mutual misunderstandings in the autistic/non-autistic social interaction. The pattern of findings between neurotypical groups by behavior interpretation abilities on an autism traits measure points to mutual misunderstandings as a clash of neurologically different social cultures. Addressing the DEP in the workplace would contribute to removing barriers to successful employment for autistic adults.

Keywords: autism, adulthood, employment, Double Empathy, social functioning

Community Brief

Why is this an important issue?

Autistic adults have difficulty finding work and staying employed. One reason for this is problems they report having with social interactions in non-autistic workplaces. Traditionally, autistic people are assumed to have trouble with social skills because they lack the ability to read what is going on in the minds of other people,

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something non-autistic people are assumed to be able to do. However, there is another theory that says the problem is not that there is something wrong with autistic people, but rather that autistic and non-autistic people do not understand each other due to being socially and neurologically different. Therefore, the problem lies not within the autistic person but in the autistic/non-autistic social interaction.

What was the purpose of this study?

This study examined the two competing ways of explaining autistic social difficulties in workplaces that are primarily non-autistic: (1) autistic people having trouble with reading the minds of other people versus (2) autistic and non-autistic people being socially different and mutually misunderstanding each other.

What did the researchers do?

With help from autistic people, we wrote a story about an autistic employee having a difficult day in a non-autistic workplace, which results in negative social interactions and an emotional reaction. We had 81 autistic and 173 non-autistic people read the story and tell us what they think is going on with the person in the story and the reason for the person's behavior.

What were the results of the study?

Autistic people who read the story were more likely to accurately understand the autistic employee in the story (50.7% accurate), compared with non-autistic people (31.2% accurate). Autistic study participants who were the best at understanding the autistic person in the story on average also reported having more characteristics of autism.

What do these findings add to what was already known?

This study adds to a growing group of studies that demonstrate that autistic people have strong social rapport and communicate well with each other and also demonstrate that non-autistic people have trouble in understanding autistic people. This study applied these previous findings to autistic social challenges in non-autistic workplaces. It contributes evidence that autistic people are not impaired socially, but rather that their social difficulties in non-autistic workplaces are due to problems within the autistic/non-autistic social interaction.

What are potential weaknesses in the study?

Because it was an online survey, people in the study could not ask questions about the story. The researchers also could not confirm participant autism diagnosis. The sample did not have enough people of different races and possibly not enough participants who routinely use the internet or social media and therefore may have been excluded from the study.

How will these findings help autistic adults now or in the future?

This work is important because it contributes to changing the way we think about why autistic people struggle to function socially in a world where the majority of people are non-autistic, including in the workplace. Rather than placing sole responsibility on autistic employees for mutual misunderstandings between autistic and non-autistic persons in the workplace, findings of this study point to shifting workplace environment and culture that understands and values autistic social differences.

Background

ADLTS ON THE autism spectrum are unemployed at high rates compared with non-autistic adults, and when compared with other disability groups.^{1–5} While unemployment rates for autistic adults are high, understanding the nature of their employment challenges is limited and few supports have been developed to address those challenges.^{6–10} Of the interventions and supports that have been developed, many focus on facilitating the transition into employment, with few focused on improving experiences for autistic employees in the workplace.^{11–14}

Difficulties autistic adults have with social interactions in non-autistic workplaces have been identified as significant barriers to employment success.^{15–18} Social functioning challenges from social interactions in non-autistic workplaces can impact many areas of work functioning including applying and interviewing for jobs,^{19–21} and communication and collaboration with coworkers and supervisors.²² Accumulated or significant negative work interactions may not only affect autistic adults' quality of life within and outside the workplace but also jeopardize employment.²² Unemployment can reduce the quality of life of an individual and negatively impact psychological health.^{23,24} Job loss is

associated with lower self-esteem, mental health concerns, deteriorated relationships, and substance misuse.²³ Explanations for autistic social challenges, including those in the workplace that may impact employment, originate from two very different perspectives: assumed autistic Theory of Mind (ToM) deficits²⁵ and the Double Empathy Problem (DEP).²⁶

Theory of Mind

Traditionally, the proposed etiology of social skills differences for autistic individuals has been their assumed lack of ToM skills.²⁵ ToM skills facilitate the accurate reading of the behaviors of others and accurately theorizing as to the thoughts, beliefs, and intentions behind behaviors.²⁵ Assumed autistic ToM deficits in social interactions have been proposed as significant barriers to employment success. ToM deficits are thought to result in many misunderstandings, negative interactions, and social anxiety for autistic employees, which can make obtaining and retaining employment much more difficult.^{15,17,27} Interventions to improve autistic social functioning often focus on correcting assumed social skills deficits, including ToM deficits, thought to lie within the autistic person.^{4,14,28,29}

Leslie described ToM skills as stemming from the ability of non-autistic children to use pretend play, which then leads to developing the ability to detect pretense in others.³⁰ Non-autistic adults may use pretense in social interactions. For example, they may say that they feel “fine” when they are unhappy and stressed so as not to appear frail or insecure to others. They may also use pretense to manipulate others for their own purposes, such as exaggerating aspects of their experience in a job interview so as to appear more competent and increase their chances of being hired. Autistic individuals are less likely to use pretense in social interactions³¹ such as to manipulate others or to present themselves in a more positive light to others.

In addition to detecting pretense, Leslie further connected the capacity for meta-representation (having a unique thought about something presented) to the ability to understand the mental states of another person and use that to predict behavior.³⁰ Therefore, ToM ability involves understanding the state of mind of another and accurately interpreting their desires, goals, and intentions.³⁰

Happé developed a measure of ToM skills in autistic children using “strange stories,”³² which have been adapted to measure ToM in autistic adults.³³ ToM is assessed by having study participants read a vignette about the actions of non-autistic characters in a story and then answer questions about the characters’ behaviors. Many strange stories include pretense in the hidden motivations of characters.³² Therefore, ToM assessment asks autistic persons to read the behaviors of non-autistic characters engaging in behaviors that are not typical of autistic persons, biasing ToM testing in favor of non-autistic ways of socially interacting. In addition, assessing ToM based on observation of only non-autistic behavior assumes that non-autistic behavior is the yardstick by which individuals are measured on their social ability, rather than equally valuing the ability to accurately read and understand autistic behavior and social functioning.

Questions have also been raised about the lack of connection between ToM assessment-measured assumed deficits

and the real-world social functioning of autistic individuals as well as autistic expressions of empathy.^{34,35} Autistic adults, academics, and researchers have raised objections to pathologizing autistic characteristics. Instead, they promote respect and appreciation for neurodiversity and equal value in the differing ways autistic people function and socialize compared with non-autistic.^{26,36–38}

The DEP

In response to the assertion that autistic individuals are impaired in their ToM abilities, in 2012, Milton has proposed that autistic social difficulties in the non-autistic world are a result of the DEP. The DEP posits that the fault for misunderstandings in autistic/non-autistic social interactions lies not within assumed social skills deficits of the autistic person (e.g., ToM), but rather within the interaction.^{26,39} Both autistic and non-autistic social partners may have difficulty “reading” the thoughts, feelings, beliefs, and intentions that accompany the behaviors of their social partner of a different neurotype.^{39,40} Meanwhile, autistic individuals socialize and communicate well with each other, similar to what often occurs among non-autistic individuals.^{26,41–43} In exploring the DEP, Milton refers to these social functioning differences between autistic and non-autistic persons as different “dispositional outlooks”²⁶ from one neurotype to the other.

Current research comparing assumed ToM deficits with the DEP

When examining social interactions through the lens of assumed autistic ToM deficits, it is expected that the poorest quality social interactions would be between matched autistic social pairs as the assumption would be that both would have ToM deficits impacting the interaction. A better quality social interaction would be between autistic and non-autistic social pairs, as the non-autistic social partner’s assumed ToM abilities would carry the interaction. The best quality interaction would be between two non-autistic social actors as it is assumed that both would have ToM ability in reading and understanding the behaviors of each other.⁴³

However, several recent studies have demonstrated that this is not the case and the present findings supporting the DEP. Observers have rated matched autistic partners as higher in social rapport than matched non-autistic partners. The lowest social rapport ratings were given for mismatched autistic/non-autistic social partners.⁴³ In a study of participants transferring story information to each other using methodology similar to the childhood game of “telephone,” groups of eight autistic participants demonstrated equally accurate story recall compared with groups of eight non-autistic participants. The poorest story recall was within the mismatched group of four autistic and four non-autistic participants.⁴²

In addition, non-autistic individuals may mistakenly believe that they accurately read the behaviors of autistic individuals, when they do not.²⁶ Non-autistic participants viewing videotapes of autistic persons reacting to social situations rated the autistic person in the video as emotionally expressive but had difficulty determining the social situation that matched their reactions.⁴⁴ One study videotaped autistic and non-autistic participants’ character animations that acted out different emotions.⁴⁵ Autistic participants who created

the animations tended to have characters with more jerky movements compared with the animations of non-autistic participants. Non-autistic observers had significantly more difficulty discerning the emotions of the autistic participants' animations compared with non-autistic participants' animations. Autistic individuals may use different movements when animating behaviors and may animate emotions differently, making interpretations more difficult for non-autistic observers.⁴⁵

Studies of the DEP have yielded informative findings supporting its premise, but research is still limited and many questions still exist including examining the DEP in different contexts.^{46,47} Social challenges in non-autistic workplaces have been implicated as significant barriers to obtaining and retaining employment for autistic adults.^{15–18} Explanations for those challenges and interventions are often from the perspective of the assumed autistic ToM deficits rather than the DEP.^{4,14,28,29} The authors are not aware of any studies that have examined the DEP in comparison to assumed ToM deficits within the context of employment. Determining the best explanation for autistic adult social challenges in non-autistic workplaces (ToM deficits vs. the DEP) has substantial implications for how this problem is addressed. Should the focus be on correcting assumed autistic deficits or on addressing mutual misunderstandings between the differing neurotypes of autistic/non-autistic?

In addition, ToM assessment has been through the lens of non-autistic social functioning as the assumed norm or standard for which all other neurodiverse ways of socially functioning are compared.^{32,33,48} As a result, ToM assessment has focused on reading and understanding non-autistic behavior rather than assessing and valuing understanding autistic behavior. The authors are not aware of any studies that have compared autistic and non-autistic interpretation of an autistic employee's behavior in the workplace.

The present study

The purpose of the present study was to examine the competing theories of ToM deficits versus the DEP by comparing autistic and non-autistic participants' ability to accurately interpret the behaviors of an autistic employee at work. Rather than assessing detecting pretense, this study examined the ability to interpret the thoughts, feelings, and intentions behind the behavior of another. The present study also sought to assess the ability to understand and interpret autistic behavior as opposed to non-autistic behavior. Accurately understanding the behavior of autistic employees is potentially very impactful on their quality of life and ability to obtain and retain employment.^{11,22,49}

A hypothesis consistent with the assumed autistic ToM deficits would expect non-autistic participants to demonstrate significantly more accurate behavior interpretation abilities toward an autistic employee compared with autistic participants. Autistic persons' assumed ToM deficits would impede their ability to accurately interpret behavior, compared with non-autistic persons who are thought to have superior ToM abilities.²⁵

A hypothesis consistent with the DEP would expect autistic participants to demonstrate significantly more accurate behavior interpretation abilities toward an autistic employee compared with non-autistic participants. According to the

DEP, the problem of autistic social challenges in non-autistic spaces lies within mutual misunderstanding in the autistic/non-autistic social interaction.²⁶ Therefore, non-autistic persons would have more difficulty interpreting the behavior of autistic persons, compared with autistic persons who often understand and connect well with each other.²⁶

Methods

Study measures

Development and scoring of the employee workplace vignette and behavior interpretation questions. For this study, a vignette was created describing a hypothetical employee in the workplace followed by behavior interpretation questions. The study vignette methodology and behavior interpretation scoring system are similar to that of Happé's strange stories ToM measure.³² In addition to ToM assessment by way of Happé's strange stories,³² vignettes and follow-up questions about attitudes and perceptions have been used to evaluate autistic characteristics in a job interview,²⁰ and in assessing perceptions of those with a mental illness.^{50,51}

To create a first draft of the vignette, a list of experiences autistic adults have reported in the workplace was created by consulting material from the following sources: Books by autistic authors,^{52–54} recorded interviews with autistic adults,⁵⁵ blog posts from the Autistic Self Advocacy Network (ASAN) website,⁵⁶ conference material on employment experiences provided by autistic adults,⁵⁷ and studies of qualitative interview data describing the lived experiences of autistic adults in the workplace.^{15,27,49,58} The vignette was written to capture many of these lived experiences, including challenges with executive functioning, routine changes, sensory sensitivities, anxiety, and confusing social interactions.^{18,27,29,59–62}

After drafting the vignette, members of the College Autism Network Virtual Association of Scholars (CANVAS),⁶³ including autistic academics and researchers, provided consultation on its content. Reviewers were asked the following questions: *Is this an authentic and realistic portrayal of an autistic employee in the workplace? Without being told, is there enough detail to infer what the hypothetical employee is experiencing?* and *What should be added, changed, or deleted?* Initial reviewers received the first draft of the vignette. It was revised and reviewed again by new CANVAS autistic reviewers through three iterations before the final version was developed. The final version of the vignette describes an employee at a fictional manufacturing company who encounters a series of workday challenges resulting in the employee becoming overwhelmed, seeking escape, and experiencing a shutting down of thought processes in response to stress.

Participants were not told that the character in the vignette was meant to portray an autistic employee, consistent with traditional ToM assessment research that does not give any information about neurodiversity or other differences. Other studies examining the perceptions of autistic behavior and the DEP have had observers of autistic behavior be blind to the autism status of the person being observed.^{43,44} Two versions of the employee workplace vignette were written with the only differences being a male versus female name for the employee (John/Julie) and the corresponding pronouns.

Participants were given the following directions: *The following is a fictional story about an employee named John (Julie) who works at a manufacturing company. Read the story carefully and then answer the questions that follow. There are no right or wrong answers, only your opinions.* After reading the vignette, study participants were asked open-ended questions about their perceptions of the thoughts, emotions, and behaviors of the employee described. The vignette and behavior interpretation questions following are included in the Supplementary Data.

Open-ended question responses were quantified using a scoring key to determine the accuracy of participants' behavior interpretation answers. This scoring system is similar to that of Happé's (1994) strange stories. Participants received a score of "2" for accurately interpreting the behavior of the hypothetical employee. To receive a score of "2," participants would need to indicate in some way by their answer that the autistic employee was (1) overwhelmed by events of the day and (2) experienced a shutdown of thought processes and was unable to respond to a question. A score of "1" was given for a partial answer in which the participant indicated only one of the two components. A score of "0" was given for an off-topic answer, "I don't know" answer, or an inaccurate understanding of the autistic employee's behavior and experience. All reviewing and scoring of open-ended questions was done blind to the neurotype (autistic/non-autistic) of the participant, as well as all other study data.

The behavior interpretation scoring key development began with the author reviewing all open-ended question content for the essential components of accurate interpretation of the autistic employee's behavior in the vignette. A preliminary draft of the answer key was created and reviewed by an occupational therapist working in the field of autism in higher education. The first author then scored all cases in the study. As a preliminary sample, a random draw of 50 cases (from the 254 cases) was scored by the third author. Scores between the first and third author were compared and reconciled with further adjustments to the scoring key, and the 50 cases were returned to the data set. The first author used the revised scoring key to again score all cases in the sample.

A random sample of 150 cases was then scored by the second author. Inter-rater reliability analyses were conducted using R version 4.2.1 (R Core Team, 2022).⁶⁴ Cohen's kappa was calculated for inter-rater reliability between the first and second authors ($\kappa=0.659$) with moderate agreement.⁶⁵ All scoring discrepancies were discussed and reconciled between the two authors, and minor edits for clarification of wording on the answer key were made. A research assistant scored the remaining 104 cases. Cohen's kappa was calculated between the first author and research assistant's scores ($\kappa=0.704$) indicating moderate agreement on the final version of the scoring key. However, all scoring discrepancies between the first author and research assistant were discussed and reconciled to obtain full consensus on scoring between raters. The final scoring system is provided in the Supplementary Data.

Measure of autistic traits and self-reported autism status. Participants completed the Ritvo Autism and Asperger Diagnostic Scale 14-item version (RAADS-14).⁶⁶ The RAADS-14 is an abridged version of the 80-item Ritvo Autism and Asperger Diagnostic Scale-Revised (RAADS-R).

The RAADS-R is a self-report questionnaire originally designed as a tool to assist with screening and diagnosis of autism.⁶⁷ Given the length of the RAADS-R, the RAADS-14 was designed as a more feasible measure while still retaining the psychometric properties of the original RAADS-R.⁶⁶ For the purposes of this study, the RAADS-14 served as an additional measure of autism traits and convergent validity for the participants self-reporting autism diagnosis or autism spectrum identification.

Previous psychometric analysis indicated the RAADS-14 has three subscales: "mentalizing deficits," "social anxiety," and "sensory reactivity."⁶⁶ A Confirmatory Factor Analysis (CFA) to confirm the three-factor structure was conducted using R version 4.2.1 (R Core Team, 2022), the lavaan R package.⁶⁸ Robust Full Information Maximum Likelihood (FIML) estimation was used to account for missing data. Initial model fit was marginally adequate based on fit indices [$\chi^2(\text{scaled})=129.8$, degrees of freedom (df)=74, comparative fit index (CFI)=0.95, Tucker Lewis index (TLI)=0.94, root mean square error of approximation ($RMSEA$)=0.061, standardized root mean square residual ($SRMR$)=0.044]. Examination of R^2 and standardized estimate values of items led to an iterative process of removing item 6 ($R^2=0.254$, *I can chat and make small talk with people—reverse scored*) and then item 1 ($R^2=0.295$, *It is difficult for me to understand how other people are feeling when we are talking*). The resulting three-factor CFA of the remaining 12 items of the RAADS-14 demonstrated significant improvement in model fit [$\Delta\chi^2(\text{scaled})=50.1$, $p<0.001$] with very good model fit based on indices [$\chi^2(\text{scaled})=79.7$, $df=51$, $CFI=0.97$, $TLI=0.96$, $RMSEA=0.053$, $SRMR=0.038$].

Analysis of missing data for the RAADS-12-item version indicated a nonsignificant value for Little's missing completely at random⁶⁹ ($\chi^2=87.5$, $p=0.744$). However, 12 cases with various missing items were imputed so that all cases could be included in analysis. Multiple Imputation (MI) was conducted on the RAADS-12-item version using R version 4.2.1 (R Core Team, 2022) and Predictive Mean Matching⁷⁰ from the *mice* (Multivariate Imputation by Chained Equation) R package.⁷¹ A pooled data set was created and used for analysis from 5 imputed data sets generated by 100 iterations each.

After completing the RAADS-14, participants were asked the following autism status questions: "Do you have an autism diagnosis?" and "Do you self-identify as autistic or a person on the autism spectrum?"

Participants

To balance the size of the autistic/non-autistic groups sufficiently for data analysis, recruitment of participants was conducted in two phases. Autistic participants were recruited first through advertisement within a listserv of scholars studying autism in higher education,⁶³ the University Center for Excellence in Developmental Disabilities, campus-wide news feed of our large public mid-western urban university, and through social media. The second phase of recruitment targeted the general population through the campus-wide news feed and social media.

Descriptive statistics of the sample including chi-square analyses, t -tests, and analyses of variance (ANOVAs) were

conducted using R version 4.2.1 (R Core Team, 2022). To examine possible sample differences in demographics and RAADS-12-item scale scores, the following analyses were conducted: Bivariate chi-square analyses were used to look for differences in frequencies of categorical demographic variables by neurotype (autistic/non-autistic). Independent samples *t*-tests were used to compare neurotype (autistic/non-autistic) group means in age and RAADS-12-item scores.

The sample consists of 254 participants, with 81 (32.2%) self-reported autistic and 173 (67.8%) self-reported non-autistic adults. The majority of the sample (60.6%) had been employed 75%–100% of their adult life, with only one person reporting never being employed since turning 18. Non-autistic participants were significantly more likely to be employed 50% of the time or more in adulthood (81.5%) compared with autistic participants (70.4%) ($\chi^2 = 3.98, p = 0.046$).

The mean age of the sample was 34.5 years (standard deviation [*SD*] = 13.5), with ages ranging from 18 to 73 years. Non-autistic participants were significantly older ($M = 36.5, SD = 14.6$) than autistic participants ($M = 30.5, SD = 10.1$) ($t = 3.64, p < 0.001$). Non-autistic participants were more likely to have a college or graduate/professional degree (62.9%) compared with autistic participants (44.4%) ($\chi^2 = 10.9, p = 0.012$). In addition, non-autistic participants were more likely to have incomes of 75,000 or above (61%)

compared with autistic participants (33.8%) ($\chi^2 = 20.3, p < 0.001$). Non-autistic and autistic participants did not significantly differ by race categories.

The majority of the sample identified as cisgender woman (65.7%), followed by cisgender man (23.5%). More than 10% of the sample identified as non-binary, gender non-conforming, or transgender, resulting in a statistically feasible third gender category. A significantly greater number of participants in the third gender category also identified as autistic ($\chi^2 = 28, p < 0.001$), consistent with emerging data indicating a substantial subgroup of autistic individuals who are also gender diverse.^{72–74}

Consistent with the literature of RAADS-14 scale scores distinguishing between autistic and non-autistic groups,⁶⁶ autistic participants had significantly higher mean RAADS-12-item scale scores ($M = 23.8, SD = 7.13$) compared with non-autistic participants ($M = 9.64, SD = 8.42$) ($t = -13.9, p < 0.001$). A full description of the sample by neurotype (autistic/non-autistic) is presented in Table 1. Please note that race/ethnicity categories were not mutually exclusive.

Procedure

Data from this study are from an online anonymous survey, collected from April through June 2022, using a cross-

TABLE 1. DESCRIPTION OF THE SAMPLE

	Non-autistic, n (%)	Autistic, n (%)	Total, n (%)
Gender ^a			
Cisgender man	34 (20.0)	25 (30.9)	59 (23.2)
Cisgender woman	128 (75.3)	37 (45.7)	165 (65.0)
Nonbinary/gender nonconforming/transgender/other	8 (4.7)	19 (23.5)	27 (10.8)
Race			
White	121 (69.9)	67 (82.7)	188 (74.0)
Black	17 (9.8)	4 (4.9)	21 (8.3)
Asian	13 (7.5)	4 (4.9)	17 (6.7)
Middle Eastern or North African	8 (4.6)	4 (4.9)	12 (4.7)
Latino or Hispanic	8 (4.6)	5 (6.2)	13 (5.1)
Native American/First Nation/Alaskan/Hawaiian/Pacific Islander/Other	11 (6.4)	4 (4.9)	15 (5.9)
Education level ^b			
High school graduate or less	17 (10.0)	19 (23.5)	36 (14.3)
Some college/associates degree	46 (27.1)	26 (32.1)	72 (28.3)
College degree	50 (29.4)	15 (18.5)	65 (25.6)
Graduate or professional degree	57 (33.5)	21 (25.9)	78 (30.7)
Household income (USD) ^a			
35,000 or less	17 (10.7)	21 (27.3)	38 (15.0)
35,000–74,999	45 (28.3)	30 (39.0)	75 (29.5)
75,000–124,999	45 (28.3)	17 (22.1)	62 (24.4)
125,000 or more	52 (32.7)	9 (11.7)	61 (24.0)
Amount of time employed in adulthood ^b			
50% or less	32 (18.5)	24 (29.6)	198 (78.0)
More than 50%	141 (81.5)	57 (70.4)	56 (22.0)
	Mean (SD)	Mean (SD)	Mean (SD)
Age ^a	36.5 (14.6)	30.5 (10.1)	34.5 (13.5)
RAADS-12-item scale score ^a	9.6 (8.4)	23.8 (7.1)	14.2 (10.4)

^a $p < 0.001$; ^b $p < 0.05$.

RAADS, Ritvo Autism and Asperger Diagnostic Scale; *SD*, standard deviation.

sectional post-test only comparative approach. Participants read a research information sheet and consented to participate by checking their agreement to do so to continue with the survey. This study was approved by the Wayne State University Institutional Review Board (IRB-22-02-4403).

Participants began the survey by reading the employee workplace vignette (participants were randomized to read about a male or female employee). After reading the vignette, participants answered open-ended questions asking them to explain the employee's behavior. Participants were asked to complete a measure of autism traits and were asked autism diagnosis or autistic identification questions. Participants also completed demographic questions including their education level, household income, and employment history. Participants were not informed that the vignette was meant to portray an autistic employee until a debriefing statement at the end of the survey.

Data analyses

Data analyses including chi-square, factorial ANOVA, and logistic regression were conducted using R version 4.2.1 (R Core Team, 2022). To address study hypotheses, bivariate chi-square analyses were used to examine the association between neurotype (autistic/non-autistic) and the behavior interpretation scores (0–2). A two-way factorial ANOVA was used to examine the interaction between neurotype and behavior interpretation scores on the RAADS-12-item scores. Given that significant neurotype group differences were found in some demographic data, logistic regression analysis was used to examine the independent contributions of the variance in the predictors of neurotype and demographics on the behavior interpretation scores.

Results

Associations between neurotype and behavior interpretation scores

A significantly greater proportion of autistic participants received a behavior interpretation score of 2 (accurate interpretation) (50.7%) compared with non-autistic participants (31.2%) ($\chi^2 = 18.6, p < 0.001$). While non-autistic participants had similar proportions of behavior interpretation scores spread across the three possible scores of 0–2 (31.2%–37.5%),

37% of autistic participants had a score of “0” (inaccurate interpretation), 12% of autistic participants had a score of “1” (partial answer but missing an important component), and just over half (50.1%) of autistic participants had a score of “2” (accurate interpretation). Proportions of behavior interpretation scores are presented in Figure 1.

A two-way factorial ANOVA examining the association between neurotype and behavior interpretation scores on RAADS-12-item scale scores indicated a significant interaction between neurotype and behavior interpretation scores ($F = 18.6, df = 2, p < 0.001$). Tukey's honestly significant difference *post hoc* tests were used to examine mean differences as follows: Within the non-autistic group, the mean RAADS-12-item scores for the participants with a behavior interpretation score of 0 ($M = 13.3$) were significantly higher than the mean RAADS-12-item scores of participants with a behavior interpretation score of 1 ($M = 8.9, p = 0.021$) and participants with a behavior interpretation score of 2 ($M = 6.87, p < 0.001$). In the autistic group, autistic participants with a behavior interpretation score of 0 had significantly lower RAADS-12-item mean scores ($M = 19.3$) than participants with a behavior interpretation score of 1 ($M = 25.1, p = 0.03$) and participants with a behavior interpretation score of 2 ($M = 26.8, p < 0.001$). Results of the RAADS-12 analysis are presented in Figure 2.

Logistic regression analysis with demographics

Given that only 10 autistic participants had a behavior interpretation score of 1, behavior interpretation scores were dichotomized by combining scores of “0” (inaccurate) and “1” (partial) into one group compared with scores of “2” (accurate). Significant multicollinearity between RAADS-12-item scale scores and neurotype resulted in RAADS-12-item scale scores being excluded from the model. Demographic variables other than gender were dichotomized to run a more parsimonious model and making it feasible to include a third gender category. Due to missing demographic variables, 34 cases were excluded from the regression ($N = 220$).

A simultaneous logistic regression predicting behavior interpretation scores was conducted with the following predictors: neurotype (non-autistic/autistic), education level (below college degree vs. college degree or higher), participant identifying as cisgender man versus cisgender woman, identifying as cisgender man versus nonbinary/transgender/

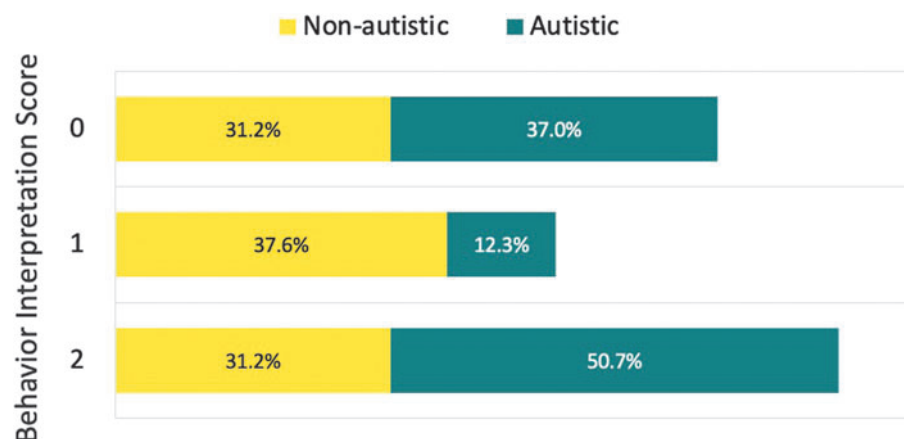


FIG. 1. Frequencies of behavior interpretation scores (0–2) by neurotype (non-autistic/autistic).

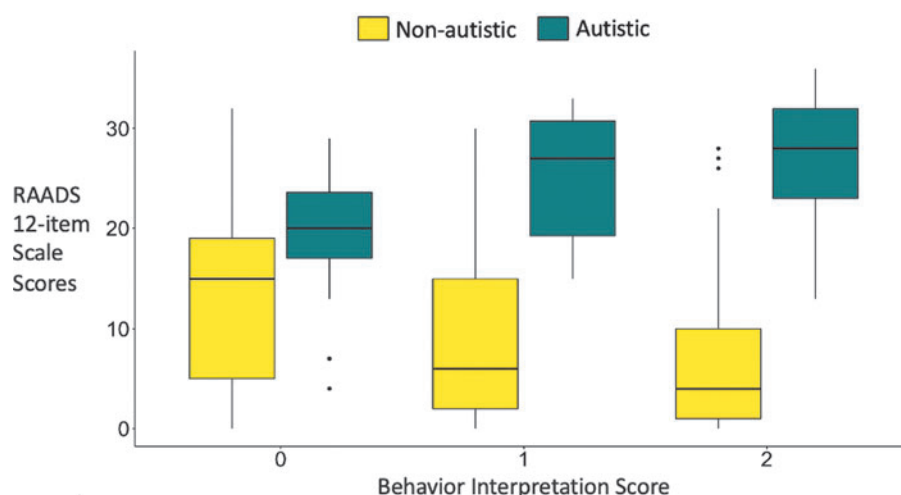


FIG. 2. Box plots of RAADS-12-item scale scores by neurotype (non-autistic/autistic). RAADS, Ritvo Autism and Asperger Diagnostic Scale.

other gender, race (White/non-White), age, total household income (74,999 or less/75,000 or more), and gender of the vignette character. The overall model was significant [$\chi^2(8)=35.3, p<0.001$], with no overdispersion in the model and a Nagelkerke's pseudo R^2 value of 0.26.

Participants who were more likely to have a behavior interpretation score of "2" were autistic (odds ratio [OR]=3.11, $p=0.003$), had a college degree or higher (OR=2.83, $p=0.005$), had higher household income (OR=2.04, $p=0.043$), identified as cisgender woman versus cisgender man (OR=3.26, $p=0.005$), and identified as nonbinary/transgender/gender nonconforming/other gender versus cisgender man (OR=8.97, $p<0.001$). Age, race, and identifying as cisgender woman versus nonbinary/transgender/gender nonconforming/other were not significant predictors in the model. Results of the logistic regression analysis are presented in Table 2.

Discussion

Results of this study are consistent with the DEP,²⁶ as opposed to the assumed ToM deficits framework of autistic social functioning.²⁵ Autistic participants were more likely to accurately interpret the hypothetical autistic employee's behavior, compared with non-autistic participants. When it

came to understanding autistic behaviors of an autistic employee and their ways of experiencing and coping in the workplace, autistic individuals had more intuitive understanding compared with non-autistic individuals. Therefore, rather than viewing poor social interactions between autistic and non-autistic in the workplace as the fault of autistic persons' assumed ToM deficits, this study supports viewing the problem as lying within the social interaction between the two different neurotypes of non-autistic and autistic.

Participants (both autistic and non-autistic) who were higher income and had more advanced education degrees were more likely to accurately interpret the behaviors of the autistic employee. Those with advanced education degrees in higher paying jobs may also have more knowledge of autism and neurodiversity, which may assist them with autistic behavior interpretation. Implications for this finding are that while initiatives supporting autistic employees in all workplaces are important, it may be especially so in workplaces that include lower wage positions or those requiring less advanced degrees. Despite autistic participants in the sample being more likely to be lower income and have fewer advanced education degrees compared with non-autistic participants, autistic participants were still more likely to accurately interpret the behavior of the autistic employee.

TABLE 2. LOGISTIC REGRESSION ANALYSIS PREDICTING BEHAVIOR INTERPRETATION SCORES OF "0/1" VERSUS "2"

Variable	Estimate	SE	OR	p	95% CI/OR
Neurotype (non-autistic/autistic)	1.13	0.387	3.11	0.003	1.48–6.79
Education level (below college/college or higher) ^a	1.04	0.367	2.83	0.005	1.39–5.93
Female vignette character ^b	0.162	0.317	1.18	0.608	0.630–2.20
Age	0.009	0.013	0.984	0.457	0.984–1.04
Cisgender woman ^c	1.18	0.425	3.26	0.005	1.46–7.84
Nonbinary/transgender/gender nonconforming/other ^c	2.19	0.634	8.97	<0.001	2.66–32.6
Race (White/non-White) ^d	−0.622	0.376	0.537	0.098	0.251–1.10
Household income ^e	0.714	0.354	2.04	0.043	1.03–4.14

Nagelkerke's pseudo $R^2=0.26, p<0.001, N=220$. Inaccurate or partially accurate interpretation of behavior=0/1, accurate interpretation of behavior=2.

^aBelow college degree=0, college degree or higher=1.

^bMale=0, female=1.

^cCisgender man is the reference group.

^dWhite=1, non-White=2.

^e74,999 or less=1; 75,000 or more=2.

CI, confidence interval; OR, odds ratio; SE, standard error.

Participants who identified as cisgender women as well as nonbinary/transgender/gender nonconforming/other were more likely to accurately interpret the behavior of the autistic employee in the vignette compared with participants who identified as cisgender men. Women may be more aware of autism and autistic ways of being and socializing as women tend to dominate professions frequently involved in supporting autistic individuals such as teaching, speech and language pathology, and occupational therapy. Women have also been considered more “psychologically minded” compared with men,⁷⁵ which may contribute to greater insight into behavior in general.

Participants who identified in the third gender category may have more appreciation for what it means to be different, to struggle socially in environments such as the workplace, and also appreciate the experience of being misunderstood by others. These experiences may contribute to their better understanding of autistic behavior. While a greater proportion of the autistic group identified as the third gender category compared with the non-autistic group, being autistic was still independently significantly associated with more accurate behavior interpretation toward the autistic employee.

Autistic individuals with the most accurate interpretation of autistic behavior also self-reported a greater number of autistic traits (RAADS-12-item version). The opposite relationship was shown for non-autistic individuals; those with the least accurate interpretations of the autistic employee’s behavior self-reported a significantly greater average number of autistic traits. Studies have demonstrated some association between self-awareness and ToM abilities.⁷⁶ Autistic individuals who are more self-aware of their autistic traits may be more intuitive regarding behaviors and mental states of an autistic other. However, this explanation is not consistent with the opposite relationship between autistic traits and behavior interpretation abilities in the non-autistic group. For the non-autistic group, self-reporting more autistic traits was associated with more difficulty reading and interpreting the behaviors of the autistic employee.

The RAADS-12-item version measure of autistic traits contains many items capturing a person’s sense of their difficulties socially functioning in a non-autistic world, such as turn taking in conversation, overfocus on details, being literal, and having difficulty with change. For autistic adults, the very characteristics that are associated with struggling socially in the non-autistic world may assist them in understanding autistic others.²⁶ In contrast, our findings indicated that non-autistic participants who reported a greater number of traits as measured by the RAADS-12-item version were less likely to understand and interpret the autistic behavior in the vignette. For those who are not autistic but report social challenges as measured by the RAADS-12-item version, even greater social disconnects may occur in both autistic and non-autistic social situations and workplaces. To address this question, future studies could examine additional varied measures of social difficulties for both groups in relation to behavior interpretation abilities of both autistic and non-autistic behavior.

Findings of this study, and others that support the DEP,^{26,41,43} indicate something qualitatively different about the ways autistic and non-autistic persons socialize and empathize.²⁶ Magnetic resonance imaging (MRI) data of brain activity has demonstrated autistic persons’ experiences of empathy particularly toward autistic characters in a story.

However, the patterns of brain activation indicating empathy were different for autistic participants compared with non-autistic participants.⁷⁷ Consistent with the DEP, the different way autistic individuals socialize may make it more difficult for them to navigate the non-autistic social world but alternatively be optimal social functioning in the autistic community.^{39,41} Autistic adults who work in companies that are “autism specific workplaces” (purposefully hire and staff their facilities with autistic employees) report fewer workplace social challenges.⁶ Autistic and non-autistic differing ways of being and socially interacting may be a clash of neurologically different social cultures.²⁶

Consistent with the findings of this study, other studies have demonstrated that autistic individuals’ ways of socializing and communicating have been misunderstood by non-autistic individuals but well understood by autistic others.^{42,43,46} Understanding autism and autistic behaviors and experiences is not the same as being autistic and experiencing life from an autistic frame of reference.⁴⁷ Evidence for the DEP points to the need for cultural humility as non-autistic persons interacting with autistic persons.³⁹ Non-autistic persons can incorrectly assume that they understand autistic persons.^{26,41,78} Such incorrect assumptions of understanding and the accompanying overtures have been experienced by autistic persons as invasive and harmful.²⁶

The workplace vignette was written to capture many lived experiences of autistic adults in the places they have been employed. Less than one third of non-autistic participants in this study correctly interpreted the behavior of the hypothetical autistic employee who was demonstrating behaviors consistent with having a difficult and overwhelming day in the workplace. The lack of understanding toward autistic employees at work as they struggle within the non-autistic community has many possible negative outcomes.

In Bury et al.’s²² recent analysis of qualitative data from autistic employees and employers, their findings highlighted that both parties tend to attribute social problems in the workplace to internal characteristics of the autistic employee. In response to workplace social challenges, it is up to the autistic employee to apologize, adjust to the environment, and avoid social situations at work to keep problems from occurring. The consequences of this approach are that the autistic employee internalizes negative views of the self.

The accumulated stress and anxiety from being blamed for what are mutual misunderstandings contribute to autistic employees’ reduced coping skills, quality of work life, work satisfaction, and mental health.²² Continued misunderstandings may cause both autistic and non-autistic coworkers to avoid each other. As the cycle of misunderstanding and avoidance evolves, further division between autistic and non-autistic individuals may result, and the marginalization of autistic adults continues.⁴¹ In addition, mutual misunderstandings and miscommunication with peers and supervisors can put autistic persons employment at risk.²²

The workplace vignette in this study also reflects power structures at play in autistic/non-autistic interactions in the workplace. There are risks for an autistic employee if they are misunderstood by management when they are struggling in a non-autistic workplace. However, autistic employees are also at a disadvantage with non-autistic coworkers as autistic employees are the neurological minority. As the neurological majority, non-autistic coworkers have the advantage of

setting the standards for norms of workplace social behavior that autistic individuals struggle to meet.^{26,47,79} The results of this study point to the need for interventions that shift workplace culture to understanding and appreciating neurodiverse social functioning and that different ways of functioning socially have equal value.^{47,79}

Findings of this study have implications for supports in the field of autism and employment. Some studies of autistic adult employment interventions have focused on addressing assumed deficits within the autistic person, including ToM deficits in social interactions,^{28,29} perpetuating the belief that autistic people are at fault for what are mutual misunderstandings. While autistic individuals have been taught specific non-autistic behaviors in response to workplace interactions or tasks, those behaviors tend not to generalize over the many nuanced and ever changing facets of non-autistic social functioning.²⁶ And, many autistic adult participants in those studies continued to remain unemployed, suggesting that specific behavioral interventions focused on fixing assumed deficits are ineffective in creating lasting success.^{9,80} For example, interventions for autistic transition-age youth tend to be based on changing discrete behaviors. Within these studies, there are not only methodological concerns but a lack of connection to the broader longer term growth of autistic adults.⁸⁰

Results of this study have implications for how we view social differences that are a barrier to autistic employment and for how we choose to support autistic adults who struggle to obtain and retain employment. Assumed impairment-focused interventions continue to place the responsibility on the autistic person rather than on the mutual misunderstandings in autistic/non-autistic social interactions. Alternatively, helpful supports would include focusing on removing barriers by modifying the environment and making workplaces more accommodating for those who are different.⁹ The growing evidence for the DEP makes imperative the need for a paradigm shift away from the focus on assumed autistic deficits and onto developing mutual understanding between autistic and non-autistic ways of being and socializing in the workplace. Adapting the workplace environment in ways that promote autistic functioning, and changing the workplace social culture to embrace neurodiverse ways of being and socializing, may contribute to longer, broader, and more successful career outcomes for autistic adults.^{9,80}

Limitations

Autistic social media recruited samples can have sampling bias with a higher proportion of females (as was the case in this study's sample), a greater number of participants with higher education degrees, and fewer participants with intellectual disabilities.⁸¹ Autistic adults who have co-occurring intellectual disabilities, who are less involved socially, less likely to use social media, less likely to be involved in online groups, less likely to volunteer for surveys, or have higher support needs may be underrepresented in this sample. There may be other biases in the sample unknown to the authors.

This study sample has an overrepresentation of White participants, especially in the autistic group (82.7%). Autism research in general lacks representation from other racial and marginalized groups.⁸² While race was not a significant predictor of behavior interpretation scores in this study, the lack of diversity in race and ethnicity in this sample excludes

the important perceptions of non-White autistic and non-autistic voices.

The use of an online anonymous survey format allowed for a larger sample size but did not allow for verification of self-reported autism status of participants. In the debate over relying on self-report in autistic adult studies rather than verifying diagnosis, researchers have noted that autism diagnosis itself is a very subjective process based on presenting behaviors and characteristics of the person, with no known universal biological markers confirming autism.⁸³ In this study, the use of the RAADS-14 measure provides some convergent validity of the sample.

In scoring the open-ended behavior interpretation questions, the researchers were not able to ask participants clarifying questions to obtain richer and more informative detail behind their answers. Scoring text only also did not allow for other nonverbal cues given by participants such as voice intonation and gestures. However, the presentation of questions online does control for any effects due to influences on participant responses from differing approaches of in-person assessment administration and any differing approaches to follow-up questions that might happen with an in-person format.

Although several autistic autism researchers consulted on the measures in this study, this work as a whole would be much more rigorous and faithful to autistic lived experiences with autistic co-creators and co-investigators.

Conclusions and Future Directions

Despite these limitations, this study contributes evidence for the DEP and demonstrates its application to autistic employment challenges. Evidence for the DEP in the workplace shifts the narrative on how we view challenges autistic adults have in finding and retaining employment. Rather than incorrectly assuming autistic social impairments, evidence for the DEP promotes viewing the problem as lying within the interaction between two different neurotypes that have differing but equally valued ways of socializing and empathizing with others.^{39,79} Addressing autistic employment challenges should involve addressing the ways the workplace environment creates and perpetuates disability by misunderstanding the behavior and challenges of autistic employees. Rather than focusing intervention on fixing what is not broken in autistic individuals, employment interventions should focus on workplace interventions that are accepting and welcoming of differences including fostering understanding and appreciation for neurodiverse ways of being and socially functioning in the world.

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Authorship Confirmation Statement

K.A.S. conceived this study, served as principal investigator, designed the study, gathered the resources for the vignette, designed and conducted the data analysis, and wrote

the article. L.A.D. significantly contributed ideas and resources for the methodology and measures, and extensively edited the article. P.D.T. helped design the scoring system and scored much of the data. P.D.T. also contributed conceptual ideas and edited the article. All authors reviewed and approved the article before submission. This article has been submitted solely to this journal and is not published, in press, or submitted elsewhere.

Author Disclosure Statement

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Supplementary Material

Supplementary Data

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